



February 20, 2026

Submitted via regulations.gov

RE: Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as part of Clinical Care

1. What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?

The biggest barrier to private sector innovation in AI for healthcare remains the persistent gap between product developers and the diversity of real-world clinical users. When developers seek out clinical advisors, they may convene panels and obtain advice from select individuals, but there remains significant variability as a user compared to other industries. There are small but meaningful nuances in how a proceduralist or surgeon views a tool compared to someone practicing in a credit-based inpatient model versus an outpatient setting. Even when advisors are consulted, blind spots may persist regarding the diversity of patients between specialties and the variability of real-world clinical workflows.

Internal regulation within health systems also remains a significant barrier. Many hospitals, including Penn Medicine, maintain AI governance boards which verify any AI tool considered for implementation in practice. Due to concerns over data ownership, privacy, intellectual property, and competitive advantage, the internal threshold can be higher than what is required for other research initiatives or clinical tools, and as a result, such issues inhibit institutional willingness to share data or participate in pilots. Today, data is increasingly

Related Docket: HHS-ONC-2026-0001-0001

Submitters:

Aanand J. Mehta; Joseph Richards; Rahil Patel; Visish M. Srinivasan, MD.

On behalf of the University of Pennsylvania Department of Neurosurgery, Cerebrovascular Division research group (NeuroVERT).

Date: February 20, 2026

Correspondence Information:

aanand.mehta@pennmedicine.upenn.edu

visish.srinivasan@pennmedicine.upenn.edu

viewed as a strategic commodity, and health systems may hesitate to engage with private companies if they perceive asymmetric value extraction or if concerns about data sharing persist.

A third barrier is workflow integration. Many AI tools appear useful when tested independently, but real impact depends entirely on how they are implemented in day-to-day clinical practice. Without plug-ins or seamless integration into PACS systems or EMRs, adoption becomes difficult. One ambient system, Dragon Copilot (Microsoft), has been integrated into EpicCharts EMR and placed into live testing at Penn Medicine, but rollout has been limited to a small number of beta users. Meanwhile, another tool through Doximity is free and widely accessible, but lacks EMR integration. Clinicians must record a conversation, permit it to transcribe to the cloud, then manually paste the output into the EMR. Manually transferring text diminishes efficiency such that traditional dictation or typing becomes equally efficient. In real-world clinical environments, if an AI tool does not function as a plug-and-play extension of existing workflows that creates more opportunity for the clinician to directly engage with the patient, adoption is unlikely, regardless of its theoretical benefits.

2. What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care?

Regulation that forces a more open-platform environment would significantly accelerate innovation.

Currently, the presence of a dominant EMR vendor (Epic) makes it difficult to innovate. When building, a company is compelled to integrate directly into Epic's ecosystem or it is otherwise effectively excluded. If an innovative company wants to expand its platform beyond Epic, it must take on additional cost burdens to build for smaller competitors (like Cerner), which is unrealistic for most startups, and therefore creates fragmented scalability. For this, strengthening 45 CFR Part 170 Subpart C, and particularly §170.315, to allow third-party AI applications to integrate without proprietary vendor barring would significantly advance the rate of innovation.

Another area that could be revisited is the rigidity of HIPAA-related restrictions as they apply to AI-enabled tools. Even when tools are HIPAA-compliant, there is institutional hesitation around data flow, transcription to the cloud, and AI processing. Currently, 45 CFR Parts 164 Subparts C and E permit the use of PHI for treatment, payment, and healthcare operations, but there exists no mention of cloud-based processing/storage, ambient transcription, and perpetual AI model refinement, rendering such regulation an institutional decision, and thus creating discontinuities between health systems regarding the rigidity of

regulations to implement. Issuing updated guidance regarding acceptable AI tool architectures and even clarifying the use of deidentified information in model training under §164.514 could reduce institutional discrepancies while preserving patient protections.

Pushing EMRs to allow an open-ended innovation platform would allow AI plug-ins to be developed in a manner that encourages portability across systems. From a regulatory perspective, such practice would reduce the structural advantage of a single dominant vendor. Without open architecture, even well-designed tools cannot scale. Interoperability mandates exist in principle, but in practice, integration still remains highly vendor-controlled. Devising certification requirements to allow plug-in flexibility and standardized API functionality while ensuring that the plug-in prioritizes patient confidentiality is therefore essential.

In addition, there are structural issues around incentive alignment and accountability. Detection-based tools may maximize sensitivity to avoid missed diagnoses, but excessive false positives create alarm fatigue and undermine clinical utility. Reimbursement structures, including CPT code development and zero-sum RVU allocation, create further disincentives for implementation of novel AI applications. Liability frameworks remain insufficiently defined when errors arise from misleading output or failed detections. To address such concerns, CMS reimbursement regulations under 42 CFR Parts 410, 414, and 415 could be revised to incentivize the use of AI-enabled services. Developing add-on payment models similar to the 42 CFR §412.87 (New Technology Add-on Payment framework) would amend current legislation mandating the redistribution of RVUs from existing services if a clinician chooses to utilize an AI tool.

3. For non-medical devices, we understand that use of AI in clinical care may raise novel legal and implementation issues that challenge existing governance and accountability structures (e.g., relating to liability, indemnification, privacy, and security). What novel legal and implementation issues exist and what role, if any, should HHS play to help address them?

With respect to novel legal and implementation issues raised by the use of non-medical device AI tools in clinical care, liability is the most immediate concern. Physicians are hesitant to use tools when they are effectively assuming all of the liability unless the tool is clearly helpful or demonstrably more likely to improve outcomes. Further, there is ambiguity around how responsibility is allocated when clinicians use AI outputs as a component of their judgement. For instance, tools “not intended for clinical *diagnosis*” like RapidAI—used in stroke workflows—remain a significant input into clinical *decision-making*. The distinction often rests on technicalities, such as viewing images on a mobile device rather than a full-resolution radiology workstation. As AI applications broaden beyond narrow use cases, exposure risk increases. Therefore, some form of

semi-indemnification framework or clarified shared-responsibility model would reduce hesitancy and encourage appropriate use without placing the full legal burden on individual physicians.

In addition, invisible data use by non-device AI systems also poses another implementation issue. Whether the responsibility of AI-informed decisions resides with IT (who prioritize security and stability), clinical leadership (patient outcomes and workflow efficiency), or both. When governance structures are dominated by IT or executive leadership removed from active practice, implementation may slow. Therefore, active clinician champions who maintain substantial clinical effort are essential to balanced governance. Otherwise, decision-making risks becoming disconnected from frontline workflow realities.

Another implementation challenge is the lack of a culture of iterative deployment; there is minimal infrastructure for opt-in early adoption models. When Dragon Copilot was released to a limited group of clinicians for beta use at Penn Medicine, the approach represented a constructive shift toward iterative refinement. However, access remained restricted, and broader opt-in participation would allow more rapid real-world learning. Allowing clinicians to opt in voluntarily could facilitate innovation without system-wide disruption. HHS could support this by clarifying pathways for phased, opt-in deployment models that preserve oversight while enabling faster feedback cycles.

4. For non-medical devices, what are the most promising AI evaluation methods (pre- and post-deployment), metrics, robustness testing, and other workflow and human-centered evaluation methods for clinical care? Should HHS further support these processes? If so, which mechanisms would be most impactful (e.g., contracts, grants, cooperative agreements, and/or prize competitions)?

Because non-medical AI tools are designed to accelerate workflow rather than determine clinical management, the legal and implementation considerations differ from those of traditional diagnostic AI systems. For example, a recently introduced Penn-native tool functions as a high-level clinical snapshot. For context, when patients present with numerous comorbidities after a complex, prolonged hospital stay, it can take substantial time to understand why the patient is being seen, particularly because most charts are organized chronologically. The summary tool synthesizes that information into a concise overview. For instance, for a patient with an incidental aneurysm, it is not necessary at the moment to know uber-specific metrics (like kidney function or ejection fraction); it is sufficient to know that the patient has heart disease, kidney disease, hypertension, and that the issue prompting the visit was incidental. Such synthesis materially improves situational awareness without replacing clinical judgment.

For tools of this nature, the relevant regulatory and programmatic focus should shift toward workflow efficiency, integration, and measurable burden reduction. Appropriate metrics could include minimizing time spent in the chart, reducing chart navigation burden, and lowering the raw number of clicks required to prepare for a visit. The ratio of pre-chart review time to actual face-to-face visit time is also meaningful. A physician's primary goal remains to increase time speaking directly with the patient while background administrative tasks become autonomous.

5. How can HHS best support private sector activities (e.g., accreditation, certification, industry-driven testing, and credentialing) to promote innovative and effective AI use in clinical care?

HHS can best support private sector activities by identifying AI applications that have demonstrated a high level of clinical improvement evidence and nudging accreditation bodies and professional standards organizations to incorporate those tools into accreditation requirements as appropriate. When an AI system shows robust, objective improvements compared to the standard of care—be it through accuracy, timeliness, or workflow efficiency—it should drive quality expectations rather than remain optional or market-dependent. For example, AI-driven stroke triage tools such as RapidAI or Viz.ai LVO have improved stroke workflow metrics, including door-to-puncture times and revascularization efficiency. For comprehensive stroke centers, it would not be unreasonable for accreditation standards to require implementation of similar tools that demonstrably improve these metrics.

This approach could extend beyond stroke to other domains such as trauma triage and time-sensitive care pathways. Though, the rebuttal to this proposition is that competition on quality does not reliably drive adoption on its own. As a result, institutions may not independently invest in tools that improve performance unless external incentives or standards reinforce their value.

6. Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short? What kinds of novel AI tools would have the greatest potential to improve health care outcomes, give new insights on quality, and help reduce costs?

A highly valuable application of AI in clinical care would be AI-based querying of individual physician performance directly within the EMR. Current tools such as Epic SlicerDicer are largely manual and limited. Extracting meaningful insights from traditional databases typically requires formal research efforts, chart abstraction, and months-long studies. It should be possible to use AI-based natural language processing

queries within the EMR to analyze one's own outcomes in real time. For example, a clinician should be able to prompt and inquire about their 30-day readmission rate or mortality rate over the last year. That level of accessibility should not require hundreds of chart reviews to generate a simple performance estimate, and would significantly contribute to iterative improvement of systems.

Similarly, procedural detail extraction should be feasible using AI/NLP capabilities. For example, when reviewing a recent aneurysm flow diverter case, a clinician may want to know precisely which instruments were used. Since, all clicks are recorded, names are embedded in the chart, and instrument documentation exists within structured and unstructured fields, AI tools should be able to extract and synthesize that information. If such a process were simple and automated, more clinicians would maintain logs to iterate on technique and refine practice patterns.

7. Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles to the adoption of AI in clinical care?

A major driver in adoption decisions is cost. Institutions are often asked to make significant upfront investments in tools without clear, immediately quantifiable returns on investment. Demonstrating downstream savings, improved throughput, or reduced complication rates often requires implementation before robust data can be generated, which creates a cyclical barrier.

Regulatory burdens on companies can also indirectly influence institutional adoption. When compliance requirements are extensive, development and validation costs increase, and those costs are passed along in the form of higher pricing. Although the HHS cannot directly control internal hospital purchasing processes, federal regulatory structure can project price through its effect on vendor operating costs. If regulatory pathways are streamlined—while maintaining safety and efficacy standards—lower development costs may translate into improved value propositions for institutions performing cost-benefit reviews.

Primary administrative hurdles include internal cost justification and prolonged review cycles through governance bodies. Lack of transparency in the decision-making structure often siloes clinical end users from financial decision-makers. Clinicians may recognize workflow benefits, but final purchasing authority rests with financial and IT leadership. Addressing cost uncertainty, clarifying reimbursement alignment, and reducing regulatory-driven price escalation would meaningfully lower the primary administrative barriers to AI adoption in clinical care.

8. Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.

While this may not represent full interoperability in the bidirectional sense, sharing datasets amongst health systems while allowing multiple AI models to run in parallel would be of significant utility. A nationally-acclaimed neurosurgery program has implemented a practice where three different stroke AI systems—such as RapidAI and Viz.ai LVO—run simultaneously. From the CT scanner, imaging data is sent to three separate cloud environments, generating three independent reports or outputs. If discrepancies arise, the physician makes the final determination. This serves as a straightforward one-way data transmission that is simple enough to implement and significantly increases robustness in decision-making. If the goal is to keep the human in the loop, triplicating models provides protection against isolated AI error.

This same principle could apply beyond imaging. Transplant risk calculators or surgical risk models could be deployed in parallel, generating independent assessments for the same patient. The same approach could be applied to dynamic surgical risk modeling before stroke intervention. A feedback cycle could be created where hospitals monitor which system aligns most accurately with their cases and what their success rates are. Such benchmarking would fuel competition because vendors could be tested head-to-head under real-world conditions rather than relying solely on pre-clinical validation claims.

From a data standards perspective, this requires standardized imaging formats, structured clinical data interoperability (e.g., FHIR-based APIs), and consistent access to timestamped workflow metrics. Beyond simply data exchange, this mandates the ability to apply multiple models to the same structured and unstructured data inputs without vendor lock-in. Benchmarking tools could then measure concordance rates, time-to-treatment metrics, and outcome correlations across models.

9. What challenges within health care do patients and caregivers wish to see addressed by the adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have related to the adoption and use of AI in clinical care?

Incorporating AI into clinical care would provide the potential to raise the floor of care by helping ensure that patients receive guideline-consistent management regardless of geography or clinician experience. AI-driven decision support embedded within the EMR can provide guardrails that improve safety, particularly for less experienced clinicians, by reinforcing evidence-based pathways and reducing variability. At the same time, AI has the potential to reduce bias by overlooking certain clinical heuristics that may influence human

decision-making, though it is also vulnerable to inheriting bias if underlying datasets are biased. For this reason feedback loops and continuous monitoring is essential.

Patients have often expressed gratitude and enthusiasm that such technologies exist. Tools such as RapidAI for large vessel occlusions and small aneurysm detection have identified findings that might not have been captured on standard radiology reads. In other contexts, AI could improve access and communication in situations where, for instance, neurosurgical patients who face long referral timelines gain access to tools to streamline triage and access. Patients increasingly engage with AI independently as well. Some bring printed outputs from LLMs, which are treated similarly to information from internet searches: understanding what the patient has interpreted and then contextualizing it clinically. These encounters can be constructive and facilitate shared understanding, particularly among technically literate patients.

However, some patients experience anxiety, particularly around incidental findings, where more information does not necessarily translate into clarity. It is often the situation that information generated through AI tools may be orthogonal to clinical relevance and contribute to unnecessary concern. As AI becomes more integrated into patient-facing and clinician-facing workflows, HHS should consider frameworks that support transparency, patient education, and contextualization of AI outputs.

10. Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?

Specific areas of AI research that HHS should prioritize to accelerate adoption in clinical care should work to refine the structure of existing medical data. One of the most important areas for prioritization is radiomics, which is highly clinical compared to more general AI developments. Radiomics represents a domain where AI can extract features and predictive signals from imaging that are not readily apparent through traditional analysis.

A closely related priority is the use of natural language processing to transform unstructured narrative text into structured data. Radiology reports and operative/clinical notes contain substantial information in free-text format. NLP tools that convert these narratives into structured labels can enable automated extraction of clinical findings, facilitate dataset curation, and support the training and validation of imaging models. For example, platforms such as RapidAI or IlluminateAI use NLP to identify patients lost to follow-up by scanning the EMR for documented findings without subsequent care.

Finally, the implementation of AI medical devices has not come without significant discrepancies in care. In practice nationally, AI devices have shown higher recall rates compared to their non-AI (predecessor) counterparts. This can be attributed to the practice where many AI-enabled devices reach market through pathways that do not require new clinical trials and may be cleared by citing prior non-AI devices as predicates. Therefore, HHS should prioritize research into real-world performance monitoring, drift detection, and lifecycle management. Post-market surveillance infrastructure is essential to ensuring the safety and likelihood of clinician and patient success.

Published findings across multiple domains support prioritizing these areas. AI-enabled triage tools have demonstrated improved thrombectomy speed in stroke workflows, increased clinical trial enrollment through automated identification of eligible patients, and improved aneurysm detection. However, adoption remains limited not by lack of technical performance but by implementation barriers. One of the most consistent challenges is compensation and workflow alignment—specifically, how clinicians and institutions are reimbursed for the additional effort required to deploy and interact with AI tools. Even when AI improves throughput or outcomes, unclear reimbursement pathways and implementation burden slow adoption.